Derek Cho

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Summary

Statistically grounded ML Researcher at the intersection of reinforcement learning, LLMs, and Bayesian inference , with a strong publication record and hands-on industry experience in frontier-model training.
Experience spans **frontier LLM post-training at Apple, RL theory, robotics, and operations**, with a track record of turning fundamental insights into algorithms that improve large-scale model training.

Education

Ph.D. candidate, Statistical Science — Durham, NC
Duke University — *2023-2027 (Expected)*

M.A. in Statistics and Data Science — South Korea, Seoul
Yonsei University • 3.90/4.00 — *2023*

B.E. in Industrial Engineering & B.A. in Applied Statistics — South Korea, Seoul
Yonsei University • 4.00/4.00, First Place — *2021*

Professional Experience

Apple | Apple Frontier Model with Reinforcement Learning — Apple Foundation Models
Research Intern — *May 2026 - August 2026*

- **Proposed a RL algorithm for Apple frontier LLMs that helped the Frontier Models team meet its 2026 reasoning target.**
- Identified a fundamental stability principle for LLM reinforcement learning: aggressive clipping is unnecessary in stable regimes
- Led End-to-end training, ablations, and stability analyses; first-author manuscript in preparation (planned September 2026 submission).

Amazon | Safe LLM Fine-Tuning with Reinforcement Learning — AWS GenAI team
Data Scientist Intern — *May 2025 - August 2025*

- Designed and implemented an RL-based framework for LLM fine-tuning that maintains safety, training models on AWS GPU infrastructure.
- Built end-to-end training, evaluation, and ablation pipelines; results supported internal deployment discussions and a 10-week cycle; published at AAAI

Research Experience

Stable Reinforcement Learning with Theoretical Guarantees — Prof. Eric B. Laber
Statistical Reinforcement Learning Lab | Research Assistant — *Jan 2025 - December 2025*

- Proposed a stable average-reward reinforcement learning algorithm addressing known optimization instabilities.
- Provided a finite-time convergence guarantee in the average-reward setting; implemented and validated the approach in operations-research and robotic-control simulations using stable reinforcement learning; published at AISTATS.

Reliable Reinforcement Learning with Order Statistics — Prof. Pan Xu
Theoretical RL group | Research Assistant — *Jan 2026 - May 2026*

- Identified heavy-tailed policy-ratio distributions as a neglected source of performance degradation in reinforcement-learning updates.
- Proved an estimator-level theoretical guarantee and developed a practical algorithm that operationalizes this insight at no additional computational cost; the resulting algorithm achieves SOTA across RL benchmarks; under review at NeurIPS.

Personalized LLMs with Theoretical Guarantees — Prof. Eric B. Laber
Statistical Reinforcement Learning Lab | Research Assistant — *November 2025 - Present*

- Developed a data-efficient RLHF personalization framework for large language models that reduces costly per-user human feedback by transferring information across users via meta-learned reward modeling.
- Established a connection to PAC-Bayesian theory to improve robustness under noisy or sparse preferences and enable uncertainty-aware learning.

Efficient Bayesian Inference with Deep Generative Models — Prof. Jaewoo Park
Bayesian ML group | Research Assistant — *March 2022 – May 2024*

- Developed a novel variational inference method that reduced computational cost by 80%+ while maintaining posterior accuracy.
- Addressed limitations of the inference method by introducing surjective mapping methods with theoretical guarantees.
- Awarded Best Paper at multiple conferences; published in the Journal of Computational and Graphical Statistics (JCGS).

Bayesian Tree Models for Science — Prof. L. Catherine Brinson
Material Science Lab | Research Assistant — *March 2025 – December 2025*

- Applied and extended novel Bayesian Tree methods to quantify predictive uncertainty for perovskite materials.
- Designed active learning with Bayesian decision criteria; implemented data pipelines, delivering a 37% improvement on target metrics
- Culminated in a publication in the Journal of Materials Chemistry A.

Selected Publications (*: Equal Contribution)

Cho D.*, Kim, H*, & Laber, E.B., "Implicit Updates for Average-Reward TD Learning" | *Conference*
AISTATS • 2026
Cho D., Chang, W., & Park J., "Fast Computer Model Calibration using Annealed and Transformed Variational Inference" | *Journal*
Journal of Computational and Graphical Statistics • 2025
Cho, D., Song, H., Goshal, A., An, H., Sharlina, K., et al., "Breaking the Safety-Capability Tradeoff: Reinforcement Learning with Verifiable Rewards
Maintains Safety Guardrails in LLMs" | *Workshop*
AAAI • 2026
Cho D., Xu, Pan, "Ordered Policy Optimization" | *Under Review*
NeurIPS • 2026
Choi M., **Cho D.,** Sheridan R., Mitzi, D. B., & Brinson, L.C., "Uncertainty-aware dimensionality prediction of low-dimensional hybrid metal halides by integrating Bayesian modeling and experiments" | *Journal*
Journal of Materials Chemistry A • 2026

Projects

Duke | Optimal Logistics Using Constrained Reinforcement Learning

Research Assistant *May 2024 - December 2025*

- Developed a constrained RL approach for large-scale vehicle-routing/network decisions, achieving 32% reduction in infections using ~2% normalized cost under the stated budget constraint.
- Built an end-to-end pipeline over 100k+ vehicle movement records and 2,500+ farms, collaborating with NCSU to define objectives, constraints, and evaluation protocols.

National Institute of Environmental Research | Geostationary Satellite Algorithm

Research Assistant *May 2022 - December 2022*

- Selected as a developer of the National Institute of Environmental Research (NIER) in Korea to implement geospatial grid algorithms for the Geostationary Environmental Monitoring Spectrometer (GEMS)
- Implemented algorithms to enhance computational speed and taking uncertainty into account, currently deployed on the Geostationary Satellite information processing

Awards & Honors

Duke Research Fellowships
Duke University • 2024, 2026
Winner of the Outstanding Student Paper Award
Joint Statistical Meetings, KISS • 2023
Best Paper (1st)
Brain Academic Conference, Yonsei University • 2022
Best Paper (1st)
Summer Academic Conference, Korean Statistical Society • 2022
Graduated with High Honors (Ranked 1st)
Yonsei University • 2021

Skills

Languages: Python, R, SQL
Machine Learning: veRL, PyTorch, HuggingFace, Transformers, RL, Offline RL
Systems & Tooling: AWS, GPU training, Git, NumPy, Pandas

Presentations

Efficient Personalization for LLMs via Meta-Learned Reward Modeling
- Duke Statistical Research Seminar (2026)

Safety-Capability Tradeoff: Reinforcement Learning with Verifiable Rewards
- New York Reinforcement Learning Conference (2025)
- AAAI (P-AI-FM) workshops (2026)

Fast Computer Model Calibration using Annealed and Transformed Variational Inference.
- AISC Conference (2024)
- Joint Statistical Meetings (2023)
- Brain Korea Academic Conference (2022)

Developing Geospatial Grid Algorithm for GEMS
- National Institute of Environmental Research (2022)